Thermal free–free (bremsstrahlung) emission in radiosynth

Equations, refractive ray tracing and numerical implementation

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radiosynth 2.2.0 — *From MHD to MHz* — November 2026

https://github.com/soumitrahazra/radiosynth

Abstract

This note collects every equation used by radiosynth to compute the thermal free–free (bremsstrahlung) radio emission of an MHD model of a solar or stellar corona. It gives the emission and absorption coefficients (constant Gaunt factor or Coulomb logarithm), the refraction of radio waves in the plasma and the ray equation, the image geometry for an observer at Earth or at a stellar distance, and the step-by-step numerical integration of the curved rays: predictor, step control by curvature and by distance to the critical surface, Boris rotation, parabolic reflection and the recovery of rays that overshoot the critical surface. It then describes the optically thin and the full radiative transfer, the conversion to brightness temperature and flux, the comparison with straight lines of sight, the spherically symmetric (1D) models and the analytic stellar-wind solution used for validation. Each section names the functions that implement it, so that the equations can be checked line by line against the code.

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1 Overview

At metre and decimetre wavelengths the quiet corona shines by thermal free–free emission: electrons accelerated in the Coulomb field of ions. Its opacity grows as $n_e^2 \nu^{-2} T_e^{-3/2}$, so at low frequencies the corona is optically thick high above the photosphere and the radio Sun is larger than the optical one, with a brightness temperature close to the coronal temperature. At the same frequencies the plasma refracts the waves strongly: rays bend away from dense plasma and are reflected before they reach the *critical density*, where the plasma frequency equals the observing frequency. Synthetic images of an MHD model therefore need both the radiative transfer and the curved ray paths.

radiosynth computes a free–free image in four steps:

1. set up the image plane and one ray per pixel, starting where the ray enters the model (Sect. 5);
2. trace all rays together through the model with the refractive ray tracer, following the algorithm of the BATS-R-US radio-wave imaging module (Benkevitch et al. 2010; Moschou et al. 2018) (Sects. 4–6);

3. along each ray, evaluate the emission and absorption coefficients (Sect. 3) and accumulate the intensity, optically thin or with the full transfer (Sect. 7);
4. convert to brightness temperature and flux (Sect. 8).

Straight lines of sight (`no_refraction=True`), 1D symmetric models and stellar observers use the same machinery (Sects. 9–11).

2 Notation, units and constants

The code works in SI units. Positions are in coordinate units of the model (solar radii $R_\odot = 6.96\times10^8$ m, or stellar radii R_*), converted to metres with the grid's `length_unit` $L_{\rm unit}$. In every Gaussian formula e^2 stands for $e^2/4\pi\varepsilon_0$.

Symbol	Meaning	Value / unit
ν	observing frequency	Hz
n_e, T_e	electron density and temperature of the model	$\mathrm{m^{-3}}$, K
ν_p	plasma frequency, $\nu_p^2 = e^2 n_e/(4\pi^2\varepsilon_0 m_e)$	Hz
$n_{\rm cr}$	critical density, $\nu_p(n_{\rm cr})=\nu$: $n_{\rm cr} = \pi m_e\nu^2/(e^2/4\pi\varepsilon_0)$	$1.2404\times10^{16}\,(\nu/\mathrm{GHz})^2\,\mathrm{m^{-3}}$
ε, μ	dielectric permittivity and refractive index, $\varepsilon=\mu^2=1-n_e/n_{\rm cr}$	—
$\boldsymbol{r}$, $\boldsymbol{v}$	ray position and unit direction, $\lvert\boldsymbol{v}\rvert=1$	coord. units, —
s, Δs	arc length along the ray and step	coord. units
$\Delta\ell$	length of a path segment in metres, $\Delta\ell = \Delta s\, L_{\rm unit}$	m
κ_ν	absorption coefficient	$\mathrm{m^{-1}}$
B_ν	source function (Rayleigh–Jeans), $2\nu^2 k_{\rm B}T_e/c^2$	$\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$
g, G	Gaunt factor; Coulomb-logarithm factor of Dulk (1985)	—
τ	optical depth from the observer	—
I_ν, T_b	specific intensity, brightness temperature	SI, K
F_ν	flux density at the observer ($1\,\mathrm{sfu}=10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$)	$\mathrm{W\,m^{-2}\,Hz^{-1}}$

In the code. `common/constants.py`: `E2_JM` $= e^2/4\pi\varepsilon_0$, `R_SUN`, `critical_density(f)`, `plasma_frequency(ne)`.

3 Emission and absorption coefficients

3.1 Source function

The coronal plasma is close to local thermodynamic equilibrium and $h\nu \ll k_{\rm B}T_e$, so by Kirchhoff's law the source function is the Rayleigh–Jeans intensity

$$B_\nu = \frac{2\nu^2 k_{\rm B} T_e}{c^2}. \tag{1}$$

3.2 Absorption coefficient with a constant Gaunt factor

The default coefficient is that of Moschou et al. (2018, Eqs. 15–16), used in BATS-R-US,

$$\kappa_\nu = \frac{4}{3}\sqrt{\frac{2\pi}{3}}\,\frac{\left[e^2/(4\pi\varepsilon_0)\right]^3}{c\,(k_{\rm B}T_e\,m_e)^{3/2}}\,\frac{n_e^2}{\nu^2}\,g, \qquad g = 10. \tag{2}$$

Numerically, in SI and in cgs,

$$\kappa_\nu = 1.7720\times10^{-12}\,g\,\frac{n_e^2[\mathrm{m^{-6}}]}{\nu^2 T_e^{3/2}}\ \mathrm{m^{-1}} \;=\; 1.7720\times10^{-2}\,g\,\frac{n_e^2[\mathrm{cm^{-6}}]}{\nu^2 T_e^{3/2}}\ \mathrm{cm^{-1}}. \tag{3}$$

3.3 Absorption coefficient with the Coulomb logarithm

With `gaunt="dulk"` the code uses Dulk (1985, Eq. 21),

$$\kappa_\nu = 9.78\times10^{-3}\,\frac{n_e^2}{\nu^2 T_e^{3/2}}\,G\ \ \mathrm{cm}^{-1}, \qquad G = \begin{cases} 18.2 + \ln T_e^{3/2} - \ln\nu, & T_e < 2\times10^5\,\mathrm{K}, \\ 24.5 + \ln T_e - \ln\nu, & T_e \geq 2\times10^5\,\mathrm{K}, \end{cases} \tag{4}$$

with n_e in cm^{-3}, $G \geq 1$, converted to m^{-1}. Comparing with Eq. (2), the two forms agree for an effective Gaunt factor

$$g_{\mathrm{eff}} = \frac{9.78\times10^{-3}}{1.7720\times10^{-2}}\,G = 0.552\,G \simeq \frac{\sqrt{3}}{\pi}\,G. \tag{5}$$

In the corona $G \approx 16$–21, so $g_{\mathrm{eff}} \approx 9$–$11.5$ and the constant $g = 10$ is a good approximation (Table 2).

Table 2: Absorption coefficient for $n_e = 10^9\,\mathrm{cm}^{-3}$, $T_e = 1\,\mathrm{MK}$ (`bremsstrahlung_coefficients`).

ν	κ ($g = 10$) [m^{-1}]	κ (Dulk) [m^{-1}]	g_{eff}
80 MHz	2.77×10^{-6}	3.07×10^{-6}	11.1
300 MHz	1.97×10^{-7}	2.04×10^{-7}	10.4
1 GHz	1.77×10^{-8}	1.72×10^{-8}	9.7
5 GHz	7.09×10^{-10}	6.25×10^{-10}	8.8

3.4 Refractive correction of the absorption

In a refracting medium the absorption coefficient along the ray is larger by the factor $1/\mu$, which matters close to the critical layer (Dulk 1985). It is not included in BATS-R-US and is off by default; with `kappa_over_mu=True` the code uses

$$\kappa_\nu \ \to\ \frac{\kappa_\nu}{\mu}, \qquad \mu = \sqrt{\max(\varepsilon, 10^{-6})}. \tag{6}$$

In the code. `bremsstrahlung/emission.py`: `bremsstrahlung_coefficients(ne, te, freq, gaunt)` returns (κ_ν, B_ν); T_e is floored at 1 K. `common/emission.py`: `EmissionModel("bremsstrahlung", gaunt=...)` is the callable used by the ray tracer. The $1/\mu$ factor is applied in `trace_rays` (`kappa_over_mu`).

4 Refraction in the coronal plasma

4.1 Refractive index and critical surface

In an unmagnetised plasma (the magnetic field has a negligible effect on refraction at these frequencies, $\nu \gg \nu_B$)

$$\varepsilon = \mu^2 = 1 - \frac{\nu_p^2}{\nu^2} = 1 - \frac{n_e}{n_{\mathrm{cr}}}, \qquad n_{\mathrm{cr}} = \frac{\pi m_e \nu^2}{e^2/(4\pi\varepsilon_0)} \approx 1.24\times10^{14}\left(\frac{\nu}{100\,\mathrm{MHz}}\right)^2 \mathrm{m}^{-3}. \tag{7}$$

Waves cannot propagate where $n_e > n_{\mathrm{cr}}$ ($\varepsilon < 0$). The critical surface $n_e = n_{\mathrm{cr}}$ acts as a mirror, and most rays are reflected well above it. Table 3 gives its height in the Newkirk (1961) corona, $n_e = 4.2\times10^4\times10^{4.32/r}\,\mathrm{cm}^{-3}$.

Table 3: Critical density and its height in the Newkirk corona; $r(n_{\rm cr}/4)$ is where $\nu_p = \nu/2$ (the level of harmonic plasma emission, for reference).

ν	$n_{\rm cr}$ [cm^{-3}]	$r(n_{\rm cr})$ [$R_\odot$]	$r(n_{\rm cr}/4)$ [$R_\odot$]
30 MHz	1.12×10^7	1.78	2.37
80 MHz	7.94×10^7	1.32	1.62
150 MHz	2.79×10^8	1.13	1.34
300 MHz	1.12×10^9	below the photosphere	1.13
1 GHz	1.24×10^{10}	below the photosphere	below

4.2 Ray equation

From Fermat's principle, $\delta \int \mu \, \mathrm{d}s = 0$, with the arc length s as parameter and $\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}s$ (Benkevitch et al. 2010; Moschou et al. 2018, Eq. 7),

$$\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} = \boldsymbol{v}, \qquad \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}s} = \boldsymbol{v} \times \left(\frac{\nabla\mu}{\mu} \times \boldsymbol{v} \right) = \frac{\nabla\mu}{\mu} - \boldsymbol{v} \left(\boldsymbol{v} \cdot \frac{\nabla\mu}{\mu} \right), \qquad \frac{\nabla\mu}{\mu} = \frac{\nabla\varepsilon}{2\varepsilon}, \quad \nabla\varepsilon = -\frac{\nabla n_e}{n_{\rm cr}}. \tag{8}$$

The direction only rotates, about the vector $\boldsymbol{\Omega} = (\nabla\mu/\mu) \times \boldsymbol{v}$, like the velocity of a charged particle in a magnetic field; this is why the Boris particle pusher can be used (Sect. 6.5). The curvature of the ray is

$$\kappa_{\rm c} = |\boldsymbol{\Omega}| = \frac{|\nabla_\perp \varepsilon|}{2\varepsilon}, \qquad |\nabla_\perp \varepsilon|^2 = |\nabla\varepsilon|^2 - (\nabla\varepsilon \cdot \boldsymbol{v})^2. \tag{9}$$

4.3 Spherically symmetric corona: Bouguer's invariant

If $n_e = n_e(r)$, Eq. (8) conserves the quantity (Bouguer's rule)

$$\mu(r)\, r \sin\psi = b, \tag{10}$$

where ψ is the angle between $\boldsymbol{v}$ and $\boldsymbol{r}$, and b is the impact parameter of the ray far from the Sun ($\mu = 1$). The ray turns at the radius r_0 where $\psi = 90°$,

$$\mu(r_0)\, r_0 = b, \tag{11}$$

above the critical surface for any $b > 0$. Fig. 1 shows rays at 80 MHz computed by the code; Eq. (10) is used to test the integration (Sect. 14).

5 Image geometry

5.1 Image plane and rays

The observer is at $\boldsymbol{O}$ (default $(-215, 0, 0)\, R_\odot$, the Earth on the $-x$ axis). With the unit vectors

$$\hat{\boldsymbol{n}} = \frac{\boldsymbol{O}}{|\boldsymbol{O}|}, \qquad \hat{\boldsymbol{\tau}} = \frac{\hat{\boldsymbol{u}} \times \hat{\boldsymbol{n}}}{|\hat{\boldsymbol{u}} \times \hat{\boldsymbol{n}}|}, \qquad \hat{\boldsymbol{\xi}} = \hat{\boldsymbol{n}} \times \hat{\boldsymbol{\tau}}, \tag{12}$$

where $\hat{\boldsymbol{u}}$ is the "up" direction (default $\hat{z}$), the image plane passes through the centre of the Sun. The pixel centres are

$$X_i = -\tfrac{1}{2} X_{\rm size} + \left(i + \tfrac{1}{2} \right) \Delta X, \qquad Y_j = -\tfrac{1}{2} Y_{\rm size} + \left(j + \tfrac{1}{2} \right) \Delta Y, \qquad \boldsymbol{P}_{ij} = X_i \hat{\boldsymbol{\tau}} + Y_j \hat{\boldsymbol{\xi}}, \tag{13}$$

($i = 0 \ldots n_x - 1$, $\Delta X = X_{\rm size}/n_x$) and the initial ray directions are

$$\boldsymbol{v}_0 = \frac{\boldsymbol{P}_{ij} - \boldsymbol{O}}{|\boldsymbol{P}_{ij} - \boldsymbol{O}|}. \tag{14}$$

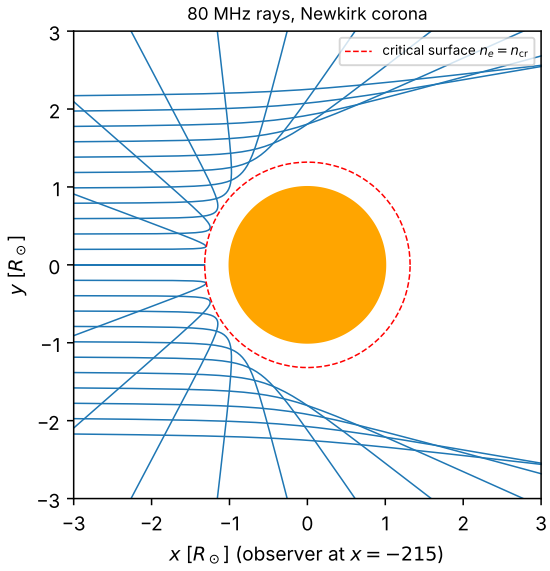


Figure 1: Ray paths at 80 MHz through a spherically symmetric Newkirk corona, computed with `rs.trace_paths`. The rays come from the observer on the left, bend away from the dense plasma and are reflected above the critical surface (red, $r = 1.32\,R_\odot$).

5.2 Integration sphere and starting point

The rays are followed inside the integration sphere of radius

$$r_{\rm int} = \left\lceil \max\left(\sqrt{(X_{\rm size}/2)^2 + (Y_{\rm size}/2)^2} + 1,\ 5 \right) \right\rceil \tag{15}$$

(as in BATS-R-US; the keyword `r_integration` overrides it). Each ray starts at the *later* of two points: where it enters the integration sphere and where it enters the model domain. The vacuum in front contributes nothing, and this matters when the model box is smaller than the sphere. Rays that miss the domain are flagged as `MISSED`.

5.3 Distant observers

For a star at distance D (`distance_pc`) the rays are parallel to $-\hat{\boldsymbol{n}}$. The ray of pixel (i, j) passes through $\boldsymbol{P}_{ij}$ and starts outside the model at $\boldsymbol{P}_{ij} + R_{\rm far}\hat{\boldsymbol{n}}$ with $R_{\rm far} = 4\max(r_{\rm int}, r_{\rm outer}, \frac{1}{2}\text{diagonal})$. The distance in coordinate units is $D = d_{\rm pc} \times 3.0857 \times 10^{16}\,\text{m}/L_{\rm unit}$.

In the code. `common/image.py`: `image_geometry` ($\hat{\boldsymbol{n}}, \hat{\boldsymbol{\tau}}, \hat{\boldsymbol{\xi}}$, pixel centres, directions; `parallel=True` for stars), `start_points` (first point inside both the sphere and the domain, status `MISSED`); `make_radio_image` sets $r_{\rm int}$ and $R_{\rm far}$.

6 Numerical integration of the rays

All active rays advance together: every quantity below is an array over the rays, and the decisions are boolean masks. The model supplies, at any point, n_e, ∇n_e (computed once on the grid with second-order finite differences and interpolated trilinearly), T_e and the local cell size $\Delta_{\rm cell}$. One step from $(\boldsymbol{r}_0, \boldsymbol{v}_0)$ with step Δs consists of the following.

6.1 Predictor and termination tests

$$\boldsymbol{r}_{1/2} = \boldsymbol{r}_0 + \boldsymbol{v}_0 \frac{\Delta s}{2}, \tag{16}$$

where n_e, ∇n_e, T_e and $\Delta_{\rm cell}$ are sampled. The ray stops if

- $|\boldsymbol{r}_{1/2}| < r_{\rm body}$ — it has reached the photosphere (`HIT_BODY`); the remaining straight piece to the surface, of length $t = -b - \sqrt{b^2 - c}$ with $b = \boldsymbol{r}_0 \cdot \boldsymbol{v}_0$, $c = r_0^2 - r_{\rm body}^2$ (limited to $[0, \Delta s]$), is still integrated with n_e capped at $n_{\rm cr}$;
- it has left the domain or the integration sphere (`ESCAPED`);
- $n_e(\boldsymbol{r}_{1/2}) \geq n_{\rm cr}$ (`OVERDENSE`; recovered in the default mode, below).

6.2 Permittivity

$$\varepsilon_{1/2} = 1 - \frac{n_e}{n_{\rm cr}}, \qquad \nabla\varepsilon = -\frac{\nabla n_e}{n_{\rm cr}}, \qquad \varepsilon_0 = \varepsilon_{1/2} - (\nabla\varepsilon \cdot \boldsymbol{v}_0)\,\frac{\Delta s}{2}, \tag{17}$$

the last being the back-extrapolated permittivity at the start of the step (Benkevitch et al. 2010, Eq. 34). If $\varepsilon_0 \leq 0$ the ray has crossed the critical surface (`NEG_EPS`).

6.3 Recovery of rays that cross the critical surface

In the default mode such rays are not discarded (Benkevitch et al. 2010, Sect. 5.2; Moschou et al. 2018, Sect. 2.2):

1. if $\Delta s \geq 4\Delta s_{\rm min}$, the step is retried from the same point with $\Delta s/4$;
2. otherwise, if the ray moves into the gradient ($\nabla\varepsilon \cdot \boldsymbol{v} < 0$), its direction is reflected specularly (Snell's law),
$$\boldsymbol{v} \;\leftarrow\; \boldsymbol{v} - 2(\boldsymbol{v}\cdot\hat{\boldsymbol{g}})\,\hat{\boldsymbol{g}}, \qquad \hat{\boldsymbol{g}} = \frac{\nabla\varepsilon}{|\nabla\varepsilon|}, \tag{18}$$
the step is reset to at least $\Delta s_{\rm step,min}$ and the steep flag is cleared.

$\Delta s_{\rm min} = 10^{-6}$ (`min_ds`) and $\Delta s_{\rm step,min} = 10^{-2}$ (`step_min`) in coordinate units.

6.4 Precision control

For rays that are not marked *steep*, two tests decide whether the step is accepted.

(a) Curvature. The squared turning angle over half a step is, from Eq. (9) with $\varepsilon \to \varepsilon_0$,

$$\left(\kappa_{\rm c}\frac{\Delta s}{2}\right)^2 = \left(\frac{\Delta s}{4\varepsilon_0}\right)^2 \left(|\nabla\varepsilon|^2 - (\nabla\varepsilon\cdot\boldsymbol{v}_0)^2\right). \tag{19}$$

If it is $\geq \mathrm{Tol}^2$ (`tolerance` $= 10^{-2}$, about 50 steps per radian), the step is redone with

$$\Delta s \;\leftarrow\; \frac{0.99\,\Delta s}{2\sqrt{(\kappa_{\rm c}\Delta s/2)^2/\mathrm{Tol}^2}}, \tag{20}$$

not below $\Delta s_{\rm min}$ in the default mode.

(b) Approach to the critical surface. If a linear prediction puts $\varepsilon = 0$ within about 1.5 steps (Benkevitch et al. 2010, Eq. 33),

$$3\,(\nabla\varepsilon\cdot\boldsymbol{v}_0)\,\frac{\Delta s}{2} \leq -\varepsilon_0, \tag{21}$$

the ray is marked steep, the distance to the critical surface $d_{\rm cr} = \varepsilon_0/|\nabla\varepsilon|$ is stored, and the step is redone with $\Delta s \leftarrow \max(\frac{1}{2}\mathrm{Tol}\,d_{\rm cr},\ \Delta s_{\rm step,min})$.

In both cases the ray stays at $\boldsymbol{r}_0$ and tries again in the next iteration.

6.5 Advance: Boris step

For an accepted step the direction is rotated with the Boris (CYLRAD) scheme (Benkevitch et al. 2010, Eq. 24). With

$$\boldsymbol{a} = \frac{\nabla\varepsilon}{2\varepsilon_{1/2}}\,\frac{\Delta s}{2} = \frac{\nabla\mu}{\mu}\,\frac{\Delta s}{2}, \tag{22}$$

$$\begin{aligned} \boldsymbol{\Omega}_0 &= \boldsymbol{a}\times\boldsymbol{v}_0, & \boldsymbol{v}' &= \boldsymbol{v}_0 + \boldsymbol{v}_0\times\boldsymbol{\Omega}_0,\\ \boldsymbol{\Omega}_{1/2} &= \boldsymbol{a}\times\boldsymbol{v}', & \boldsymbol{v}'' &= \boldsymbol{v}_0 + \boldsymbol{v}_0\times\boldsymbol{\Omega}_{1/2},\\ \boldsymbol{v}_1 &= \boldsymbol{v}_0 + \frac{2}{1+\Omega_{1/2}^2}\,\boldsymbol{v}''\times\boldsymbol{\Omega}_{1/2}, & \boldsymbol{r}_1 &= \boldsymbol{r}_{1/2} + \boldsymbol{v}_1\frac{\Delta s}{2}. \end{aligned} \tag{23}$$

The rotation keeps $|\boldsymbol{v}| = 1$ exactly and is second-order accurate. The segment length is Δs.

6.6 Advance: parabolic switch at the critical surface

For rays that satisfy Eq. (21) even after the step was reduced, the ray is close to its turning point. If $\nabla\varepsilon$ is constant over the step the path is a parabola. With $L = 1/|\nabla\varepsilon|$ and α the angle between $\boldsymbol{v}_0$ and $-\nabla\varepsilon$,

$$\begin{aligned} L\cos\alpha &= -\frac{\nabla\varepsilon\cdot\boldsymbol{v}_0}{|\nabla\varepsilon|^2}, & \boldsymbol{v}_\parallel &= -L\cos\alpha\,\nabla\varepsilon,\\ \boldsymbol{v}_1 &= \boldsymbol{v}_0 - 2\boldsymbol{v}_\parallel, & \Delta\boldsymbol{r}_\perp &= 4\,(\boldsymbol{v}_0 - \boldsymbol{v}_\parallel)\,L\cos\alpha\,\varepsilon_0,\\ \boldsymbol{r}_1 &= \boldsymbol{r}_0 + \Delta\boldsymbol{r}_\perp, & \Delta\boldsymbol{r}_\parallel &= \varepsilon_0\,(L\cos\alpha)^2\,\nabla\varepsilon, \end{aligned} \tag{24}$$

and the emission of the arc is counted over the length

$$\Delta s_{\rm arc} = \sqrt{|2\Delta\boldsymbol{r}_\parallel|^2 + |\Delta\boldsymbol{r}_\perp|^2}. \tag{25}$$

The ray jumps to the symmetric point on the other branch of the parabola, with its normal velocity component reversed.

6.7 Clipping at the photosphere

If the new point $\boldsymbol{r}_1$ lies inside $r_{\rm body}$, only the fraction $t \in [0,1]$ of the segment outside the body is counted, from the first root of $|\boldsymbol{r}_0 + t(\boldsymbol{r}_1 - \boldsymbol{r}_0)| = r_{\rm body}$; the ray ends (`HIT_BODY`).

6.8 Step regrowth

After an accepted step, for gentle rays and for steep rays that have moved away from the critical surface ($\varepsilon_0 > d_{\rm cr}|\nabla\varepsilon|$), the steep flag is cleared and

$$\Delta s \;\leftarrow\; \max\!\left(2 - \frac{\Delta s}{\Delta s_{\rm new}},\,1\right)\min(\Delta s_{\rm new}, \Delta s), \qquad \Delta s_{\rm new} = f_{\rm step}\,\Delta_{\rm cell}, \tag{26}$$

which roughly doubles small steps and saturates smoothly at $\Delta s_{\rm new}$ ($f_{\rm step}$ = `step_factor`, default 1). The first step of a ray is $f_{\rm step}\Delta_{\rm cell}$ at its starting point (or `ds_init`).

6.9 The original BATS-R-US logic

With `strict_batsrus=True` the code reproduces the step logic of the BATS-R-US module exactly. It differs from the default in two places:

1. BATS-R-US also uses the parabolic switch whenever $\Delta s < \Delta s_{\mathrm{step,min}}$ and $\nabla\varepsilon \cdot \boldsymbol{v} < 0$. This condition is reached after the curvature control has shrunk the step, so it fires on rays far from the critical surface, which then jump along a long, wrong parabola. The default mode uses the switch only under Eq. (21).
2. BATS-R-US discards rays that land in over-dense plasma or have $\varepsilon_0 \le 0$; the default mode recovers them (step back and Snell reflection).

The default mode also has the lower bound $\Delta s_{\min}$ in the curvature control.

6.10 Ray status

Every ray ends with one of the codes `ESCAPED` (left the domain or the sphere), `HIT_BODY`, `OVERDENSE`, `NEG_EPS` (the last two mainly in strict mode or without refraction; in the default mode only rays that cannot be reflected end this way), `OPAQUE` ($\tau > \tau_{\max}$, full transfer), `MAX_STEPS` (safety limit, 200 000 iterations) or `MISSED`; they are stored per pixel in `img.status`.

In the code. `common/raytrace.py`, function `trace_rays`, one subroutine per stage: `_predictor`, `_terminations`, `_integrate_to_body`, `_dielectric` (Eq. 17), `_snell_fallback`, `_check_step` (Sect. 6.4), `_advance` with `_boris_step` and `_parabolic_step`, `_clip_at_body`, `_add_emission` (Sect. 7) and `_new_step_size`. Keywords: `tolerance`, `step_min`, `min_ds`, `step_factor`, `ds_init`, `strict_batsrus`, `max_steps`. The gradient ∇n_e is computed in `common/grid.py` (`np.gradient`, second order).

7 Radiative transfer along the ray

For each accepted segment, κ_ν and B_ν are evaluated at the mid-point $\boldsymbol{r}_{1/2}$ and the segment length in metres is $\Delta\ell = \Delta s\, f_{\mathrm{clip}}\, L_{\mathrm{unit}}$ (with the arc length for a parabolic step and the clipping fraction at the photosphere).

7.1 Optically thin (default, as in BATS-R-US)

$$I_\nu = \sum_{\text{segments}} \kappa_\nu B_\nu \, \Delta\ell. \tag{27}$$

This is what BATS-R-US and Moschou et al. (2018, Eq. 14) use. It neglects absorption, so in the dense low corona it can give $T_b > T_e$ (Fig. 2, right).

7.2 Full transfer

With `transfer="full"` the code solves $\mathrm{d}I/\mathrm{d}s = \kappa(B - I)$ exactly for constant κ and B on each segment, integrating from the observer into the medium:

$$I_\nu = \sum_{\text{segments}} B_\nu \, e^{-\tau} \left(1 - e^{-\Delta\tau}\right), \qquad \Delta\tau = \kappa_\nu \Delta\ell, \qquad \tau \leftarrow \tau + \Delta\tau, \tag{28}$$

where τ is the optical depth between the observer and the start of the segment ($1 - e^{-x}$ is evaluated as `-expm1(-x)`). This gives $T_b \le T_e$ in isothermal plasma and the optical depth τ of every pixel. Once $\tau > \tau_{\max}$ (default 30, $e^{-30} \approx 10^{-13}$) nothing more can be seen and the ray stops (`OPAQUE`). The full transfer is recommended for physically meaningful brightness temperatures below about 1 GHz.

Neither mode adds emission from the photosphere behind a ray that reaches r_{body}.

In the code. `common/raytrace.py`: `_add_emission(intensity, tau, idx, kappa, B, dl, transfer)` (Eqs. 27 and 28); `tau_max` in `trace_rays`.

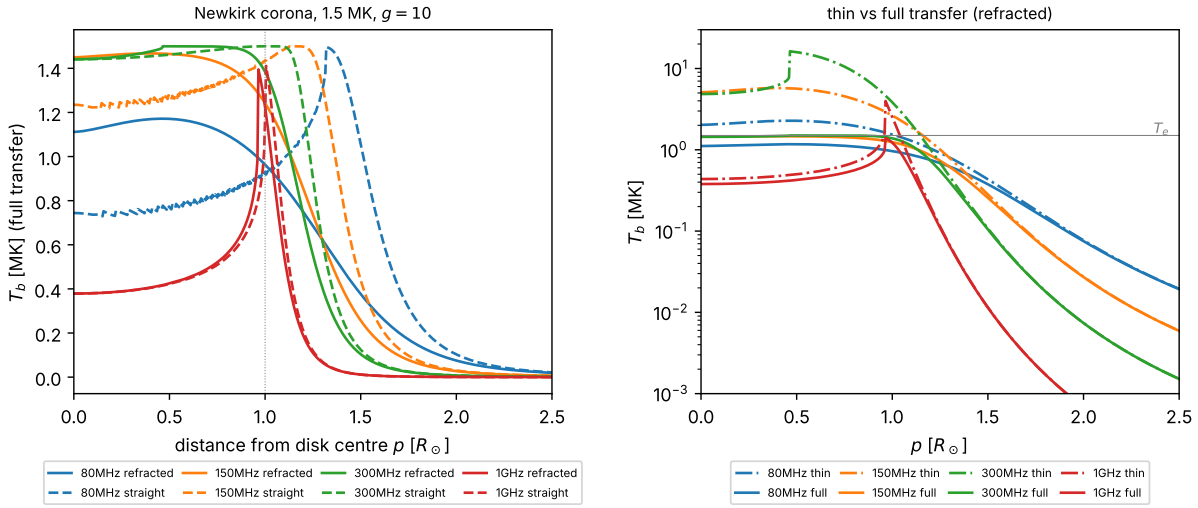


Figure 2: Brightness temperature against distance from the disk centre for a spherically symmetric Newkirk corona with $T_e = 1.5\,\mathrm{MK}$ and $g = 10$, computed with `rs.make_symmetric_image`. Left: full transfer, refracted rays (solid) and straight lines of sight (dashed). At 80 and 150 MHz refraction makes the radio Sun smaller (the limb falls off closer to the disk; the fluxes are 3.1 and 11.7 sfu instead of 4.1 and 14.4 sfu) but its centre brighter: straight lines stop at the critical surface after an optical depth below one, while refracted rays turn above it and travel longer through the dense plasma. At 1 GHz the two agree. Right: refracted rays, thin (dash-dotted) and full (solid) transfer; the thin sum exceeds T_e where the corona is optically thick.

8 Brightness temperature and flux

$$T_b = \frac{c^2}{2k_\mathrm{B}\nu^2}\, I_\nu, \qquad F_\nu = \sum_{\text{pixels}} I_\nu\, \Delta\Omega, \qquad \Delta\Omega = \frac{\Delta X\, \Delta Y}{D^2}, \tag{29}$$

with $D = |\boldsymbol{O}|$ (or the stellar distance) and all lengths in the same unit. The flux is also given in sfu ($10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$) and Jy ($10^{-26}$).

In the code. `common/image.py`, class `RadioImage`: `intensity`, `tau`, `status`, `brightness_temperature`, `pixel_solid_angle`, `flux_density`, `summary()`, `save_npz`, `save_fits`, `plot`.

9 Refracted rays versus straight lines of sight

With `no_refraction=True` the rays are straight lines: the step control and the reflection are skipped, the direction is never changed and the step regrows to the cell size. The rays still stop where $n_e \geq n_\mathrm{cr}$, since the waves cannot propagate there. The difference between the two images measures the effect of refraction (Fig. 2, left). At low frequencies the radio Sun is smaller with refraction, because rays passing near the limb are bent away from the dense plasma and see less of it. Near the disk centre the opposite can happen: a straight line stops at the critical surface, possibly before it becomes optically thick, whereas a refracted ray turns above the critical surface and accumulates more optical depth near its turning point, so the centre can be brighter with refraction. At high frequencies, where $n_e \ll n_\mathrm{cr}$, the two agree.

`rs.compare_refraction(grid, freqs, ...)` computes both images with identical options and returns a dictionary with the two images (`refracted`, `straight`), the flux ratio refracted/straight per frequency (`flux_ratio`) and, if `path` is given, a figure of both images and their ratio (`-compare-refraction` on the command line).

10 Spherically symmetric (1D) models

A 1D model gives $n_e(r)$ and $T_e(r)$ (`rs.grid_from_profile`, `rs.read_profile`, or 1D spherical PLUTO output). Its image depends only on the distance p from the disk centre, and the problem is symmetric about the observer–Sun axis also with refraction and for an observer at finite distance. `rs.make_symmetric_image` therefore traces a single row of rays, $p_k = k\,\Delta p$, $k = 0 \ldots n_p$ ($\Delta p = p_{\max}/n_p$, default $n_p = 1000$), rotates the result into a 2D image, and integrates the flux over the whole disk $p \le p_{\max}$ exactly:

$$F_\nu = \frac{2\pi}{D^2} \int_0^{p_{\max}} I_\nu(p)\, p\, \mathrm{d}p \tag{30}$$

(trapezoidal rule). It also returns the profiles $I_\nu(p)$ and $T_b(p)$ (Fig. 2).

In the code. `common/symmetric.py`: `grid_from_profile`, `read_profile`, `make_symmetric_image` (one row of $2n_p + 1$ rays through the centre, rotation with `_rotate`, profile flux), `plot_profile`.

11 Stars and the analytic wind

For a star, coordinates are in stellar radii ($L_{\rm unit} = R_*$), the observer is at a distance in parsec and the rays are parallel (Sect. 5). The code is validated against the classic isothermal, fully ionised wind $n_e = n_0 (r_0/r)^2$ with constant g (Wright & Barlow 1975; Panagia & Felli 1975). Writing $\kappa = \kappa_1 n_e^2$, the optical depth along a straight line at impact parameter p is

$$\tau(p) = \kappa_1 n_0^2 r_0^4 \int_{-\infty}^{\infty} \frac{\mathrm{d}l}{(p^2 + l^2)^2} = \frac{a}{p^3}, \qquad a = \frac{\pi}{2}\,\kappa_1 n_0^2 r_0^4. \tag{31}$$

The flux of the infinite wind (neglecting the stellar disk) is

$$F_\nu = \frac{2\pi B_\nu}{D^2} \int_0^\infty \left(1 - e^{-a/p^3}\right) p\, \mathrm{d}p = \frac{2\pi B_\nu}{D^2}\,\frac{a^{2/3}}{3} \int_0^\infty (1 - e^{-x})\, x^{-5/3} \mathrm{d}x = \frac{\pi\,\Gamma(1/3)\, B_\nu\, a^{2/3}}{D^2}, \tag{32}$$

using $x = a/p^3$ and $\int_0^\infty (1 - e^{-x}) x^{-5/3} \mathrm{d}x = \frac{3}{2}\Gamma(\frac{1}{3})$ (integration by parts). Since $B_\nu \propto \nu^2$ and $a \propto \nu^{-2}$, $F_\nu \propto \nu^{2/3}$ (the classic $\nu^{0.6}$ once the weak frequency dependence of the Gaunt factor is included). For $p < r_0$ the line stops at the stellar surface and the code integrates the front half numerically.

In the code. `bremsstrahlung/wind.py`: `wind_optical_depth`, `wind_flux_density`; `common/stellar.py`: stellar observers, spectra, spectral indices.

12 Magnetised free–free emission

In a magnetic field the two magneto-ionic modes have different free–free absorption,

$$\kappa_\sigma = \frac{\kappa_\nu}{(1 + \sigma q)^2}, \qquad q = \min(Y|\cos\theta|, 0.9), \qquad Y = \frac{\nu_B}{\nu}, \quad \sigma = +1\ (\mathrm{o}),\ -1\ (\mathrm{x}), \tag{33}$$

each with half of B_ν, which gives $V/I = 2q/(1 + q^2) \simeq 2Y|\cos\theta|$ for a thin source. This polarised form is used by the gyroresonance and combined images, along straight rays (see the companion note on gyroresonance emission); the refractive images described here are unpolarised.

13 Assumptions and range of validity

1. **Thermal, Rayleigh–Jeans emission**, $h\nu \ll k_{\rm B} T_e$, with a constant or Coulomb-logarithm Gaunt factor; free–free emission of electrons on ions only.

2. **Geometrical optics.** The model must resolve the density scale height near the critical layer; with cells of 0.025–0.05 $R_\odot$ the accuracy is about 1 %.
3. **Refraction in an unmagnetised plasma**; no scattering by density fluctuations, which matters below about 200 MHz (Moschou et al. 2018), and no plasma (coherent) emission.
4. **No emission from the photosphere or chromosphere** behind rays that reach $r_{\rm body}$; above ~ 1 GHz the model must contain the chromosphere and transition region for realistic disk brightness.
5. **Thin transfer** is valid only where $\tau \ll 1$; use the full transfer otherwise.

14 Validation

The scripts `tests/validate.py`, `tests/test_stellar.py`, `tests/test_symmetric.py` and `tests/test_refraction_compare.py` check the implementation (Newkirk corona, $T_e = 1.5$ MK, 0.025 $R_\odot$ grid unless stated otherwise):

Test	Result
Bouguer invariant $\mu r \sin\psi$ conserved along rays (80 MHz)	1.2 % max. deviation
Turning radius vs. analytic $\mu(r_0) r_0 = b$	$4 \times 10^{-4}\, R_\odot$
Exit direction vs. `scipy solve_ivp` of Eq. (8) (150 MHz)	$\leq 0.3°$
Straight-ray intensity vs. quadrature (1, 17 GHz)	≤ 0.9 %
Full transfer: $T_b \leq T_e$	yes
Flux vs. step size (`step_factor` $1 \to 0.25$, 150 MHz)	3×10^{-4}
Stellar wind: τ per pixel vs. analytic	0.08 %
Stellar wind: flux vs. exact; analytic F_ν vs. quadrature	0.02 %; 10^{-7}
Coordinates in $R_\odot$ vs. R_*	identical
Symmetric image (one row of rays) vs. full 2D image: flux; T_b along a row	2×10^{-4}; 3×10^{-4}
1D profile grid vs. 3D Cartesian grid of the same corona	0.02–2 %
Optically thick wind from a text table: profile flux vs. exact integral	2×10^{-4}
`compare_refraction` vs. separate runs; 17 GHz refracted/straight flux	identical; $1 - 2 \times 10^{-4}$

15 Using it

```
import radiosynth as rs
grid = rs.load_pluto("out/", nout=27)        # or rs.load_mhd("box.dat")
img = rs.make_radio_image(grid, "80MHz, 150MHz, 300MHz", x_size=6, y_size=6,
                          nx=256, ny=256, transfer="full", gaunt=10.0)
img.plot("tb"); print(img.summary())
cmp = rs.compare_refraction(grid, "80MHz", transfer="full", path="refraction.png")
print(cmp["flux_ratio"])     # cmp["refracted"], cmp["straight"] are images
```

or, from the command line,

```
python -m radiosynth out/ --freq "80MHz, 150MHz" --size 6 6 --npix 256 256 \
       --transfer full --plot --out quiet_sun
```

Keyword of `make_radio_image`	Quantity in this note
`gaunt` (number or `"dulk"`; default 10)	g or G, Sect. 3
`transfer` (`"thin"`, `"full"`)	Eq. (27) or (28)
`no_refraction`	straight lines, Sect. 9
`kappa_over_mu`	κ/μ
`tau_max`	$\tau_{\max}$ (full transfer)
`tolerance`, `step_min`, `min_ds`	Tol, $\Delta s_{\mathrm{step,min}}$, $\Delta s_{\min}$
`step_factor`, `ds_init`	f_{step}, first step
`strict_batsrus`	Sect. 6.9
`observer`, `up`, `x_size`, `nx`, `r_integration`, `distance_pc`	Sect. 5
`n_workers`	parallel processes over the rays

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